Abhiram-S
Roll no. 1

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

SCHOOL OF ENGINEERING

B. TECH DEGREE SECOND SEMESTER FIRST INTERNAL EXAMINATIONS, APRIL 2024

23-200-0202B ENGINEERING CHEMISTRY

(CS/EC/IT/EEE)

(2023 Scheme)

TIME: 2 hrs **MAX. MARKS: 30**

Course Outcomes

On completion of this course the student will be able to:

CO1: Explain the basic concepts of chemical thermodynamics, and quantum chemistry.

CO5: Understand the principle, concept, working and applications of relevant technologies and comparison of results with theoretical calculations.

PART A

(Answer all questions) **(5 x 2= 10 Marks)**

Q.No.	Questions	Marks	BL	CO	PO
I (a)	Calaculate the number of degrees of freedom in the following systems: (i) An aqueous solution of NaCl (ii) Mixture of $N_2(g)$, $H_2(g)$, and $NH_3(g)$	2	L2	1	1
(b)	Derive the expression connecting internal energy and Heat capacity.	2	L2	1	1
(c)	Write a note on thermodynamics of biochemical reaction.	2	L1	5	1
(d)	Give all the possible values of quantum numbers for (i) Valence electron in sodium (Atomic No of Na: 11) (ii) Valence orbital of Titanium (Atomic No,[redacted] 22)	2	L2	5	
(e)	Discuss the characters of an acceptable wavefunction.	2	L1	1	2

PART B

(Answer the following questions choosing either the first OR second option)

(1 x 10= 10 Marks)

Q.No.	Questions	Marks	BL	CO	PO
II (a)	Imagine that an electron is continuously moving in x direction only in a box of ∞ height and length a. Develop an	7	L2	5	2

		acceptable wavefunction and determine energy states of an electron.				
	(b)	Explain Hamiltonian operator.	3	L1	5	2
		OR				
III	(a)	Write down the hydrogen atom wavefunctions interms of polar coordinates. How do the quantum numbers n, 1, and m emerge from the solution of wave equation? Write the significance of all the quantum numbers.	7	L3	5	2
	(b)	For the wavefunction $\psi = A \cos x$ for $0 < x < \pi/2$. Find the value of A.	3	L3	5	2
IV	(a)	What is Gibbs Free energy? Explain mathematically how free energy is related to pressure and temperature. Derive Gibbs-Helmholtz equation and describe the spontaneity conditions.	10	L2	1	2
		OR				
V	(a)	Explain the water system with a neat labelled phase diagram.	7	L2	1	2
	(b)	Explain the phase rule, which is applicable in the case of two component system having solid – liquid equilibrium.	3	L2	1	2

Bloom's Taxonomy Levels

L1 : 25% L2: 58.33% L3: 16.6%